MILITARY SPECIFICATION

POWER SUPPLY ASSEMBLIES FOR FASCAM ELECTRONIC ASSEMBLIES

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by drawings and provides Quality Assurance provisions for the fabrication of parts, assembly and packing of one type of power supply assembly.

2. APPLICABLE DOCUMENTS

2.1 Issues of document. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified.

SPECIFICATIONS

MILITARY

MIL-S-45743	-	Soldering, Manual Type, High Reliability Electrical And Electronic Equipment
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications For
MIL-B-48761	-	Battery Single Cell, Primary Reserve

STANDARDS

MIL-STD-105	-	Sampling Procedures And Tables For Inspection By Attributes

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-202 - Test Methods For Electronic And Electrical Component Parts

MIL-STD-1169 - Packaging, Packing And Marking For Shipment Of Inert Ammunition Components

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9287154 - Power Supply Assembly

9292241 - Power Supply Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawings (DWG.) - 9287154/9292241 and 9287159/9292221 with all requirements specified in applicable specifications and standards.

3.3 Electrical requirement

3.3.1 Non-activated voltage. Each battery cell shall be unactivated and measure less than .005 volts.

3.3.2 Shorting bar resistance. The shorting bar and connector assembly shall have less than 0.8 ohm resistance measured between the terminals.

3.4 Battery lead weld integrity. The battery assembly lead welds shall not fail or show any evidence of tearing or breaking when the leads are subjected to five (5) pound plus two (2) pounds pull.

3.5 Soldering

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3.5.1 Shorting bar and connector assembly solder workmanship. Soldering of the shorting bar and connector assembly shall be in accordance with MIL-S-45743 and Appendix herein.

3.5.2 Solderability. The power supply assemblies ribbon leads shall comply with solderability requirements of MIL-S-45743.

3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner and all manufacturing processing and assembly operations shall be correctly performed. All parts shall be free from chips, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the part be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the parts/assemblies listed below in the quantities indicated.

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Part Description	Drawing	Quantity
Power Supply Assembly	9287154/9292241	25
Shorting Bar, Ceramic	9292157	25
Shorting Bar and Connector Assembly	9292158	25
Molding, Power Supply	9287159/9292221	25

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE: Power Supply Assembly		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Molding, Power Supply (Dwg. 9287159/9292221)				
	Examination for defects	25		3.2	4.4.2.1/4.4.2.6
	Shorting Bar, Ceramic (Dwg. 9292157)				
	Examination for defects	25		3.2	4.4.2.3
	Shorting Bar and Connector Assembly (Dwg. 9292158)				
	Examination for defects	25		3.2	4.4.2.4
	Solderability	25		3.5.1	4.4.2.4
	Power Supply Assembly (Dwg. 9287154/9292241)				
	Examination for defects	25		3.2	4.4.2.5
	Pull test	25		3.4	4.5.3
	Solderability	10		3.5.2	4.5.5

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of Power Supply Assemblies shall contain batteries bearing the same lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.1	Power Supply Molding		9287159
			NEXT HIGHER ASSEMBLY 9287154

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	I.D. of middle hole (Sect AA)		0.40%	3.2	Gage/4.4.5
102	O.D. of bottom post (max)		0.40%	3.2	Gage/4.4.5
103	True position of post		0.40%	3.2	Gage/4.4.5
104	Width of bridge		0.40%	3.2	Gage/4.4.5
105	True position of bridge width		0.40%	3.2	Gage/4.4.5
106	I.D. of largest center hole		0.40%	3.2	Gage/4.4.5
107	True position of largest hole		0.40%	3.2	Gage/4.4.5
108	Height from bottom to top of bridge		0.40%	3.2	Gage/4.4.5
109	Material		0.40%	3.2	Certificate/6.2
110	Distance from center line to edge of shorting bar hole (Section B-B)		0.40%	3.2	Gage/4.4.5
111	Length of bottom post		0.40%	3.2	Gage/4.4.5
112	I.D. of smallest hole		0.40%	3.2	Gage/4.4.5
Minor					
201	Length from bridge to closed end (2 places)		0.65%	3.2	Gage/4.4.5
202	True position of inside width of cell cavity (2 places)		0.65%	3.2	Gage/4.4.5
203	Width of shorting bar hole (Sect. B-B)		0.65%	3.2	Gage/4.4.5
204	Length of shorting bar slot		0.65%	3.2	Gage/4.4.5
205	Width of cell cavity (2 places)		0.65%	3.2	Gage/4.4.5

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Power Supply Molding				9287159
					NEXT HIGHER ASSEMBLY 9287154
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
206	Depth of shorting bar hole (Sect. B-B)		0.65%	3.2	Gage/4.4.5
207	Distance from center line to edge of shorting bar hole		0.65%	3.2	Gage/4.4.5
208	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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MIL-P-63421(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Support Foam				9292197
					NEXT HIGHER ASSEMBLY
					9287154/9292241
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor 201	Adhesive missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Shorting Bar, Ceramic	1 OF 1	9292157
			NEXT HIGHER ASSEMBLY 9287158

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Solderability	15	Note 2	3.2	Note 1
Minor					
201	Total height		0.65%	3.2	Gage
202	Total length		0.65%	3.2	Gage
203	Dielectric glaze missing or incomplete		0.65%	3.2	Visual
204	Width of conductive strip, 3 places		0.65%	3.2	Gage
205	Solder coating outside area "A" incorrect		0.65%	3.2	Visual
206	Length of neck, 4 places		0.65%	3.2	Gage
207	Length to outside of two conductive legs		0.65%	3.2	Gage
208	Depth to neck, 4 places		0.65%	3.2	Gage
209	Thickness of solder pad (2 places)		0.65%	3.2	Gage
210	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Per MIL-S-45743.
(2) Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Shorting Bar and Connector Assembly		9292158
			NEXT HIGHER ASSEMBLY 9287154/9292241

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship incorrect		0.40%	3.5.1	4.5.4
102	Evidence of cracks or chips		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Power Supply Assembly		9292154/9292241
			NEXT HIGHER ASSEMBLY
			9287138/9292223/9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pull test (weld integrity)		100%	3.4	4.5.3
Major					
101	Non-activated voltage (each cell)		Note 2	3.3.1	Note 1, 4.5.1
102	Weld integrity (fractured weld or hole in case)		Note 2	3.2	Visual
103	Battery cell(s) not bonded to mold		0.40%	3.2	Visual
104	Shorting bar(s) support foam missing		0.40%	3.2	Visual
105	Shorting bar assemblies not properly seated in cavity		0.40%	3.2	Visual
106	Air gap present between support foam and cavity		0.40%	3.2	Visual
107	Length of lead(s) extending below mold (min)		0.40%	3.2	Gage
108	Resistance of shorting bar(s)		0.40%	3.3.2	4.5.2
109	Solderability, 4 cell leads	15	0-1	3.5.2	4.5.5
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Use 10^7 ohm input impedance voltmeter.
(2) Sample per MIL-STD-105D, Level II. Reject for one (1) or more failures.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Power Supply Molding	1 OF 2	9292221
			NEXT HIGHER ASSEMBLY 9292241

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	*			
Major					
101	I.D. of small center hole		0.40%	3.2	Gage/4.4.5
102	O.D. of bottom post (max)		0.40%	3.2	Gage/4.4.5
103	True position of O.D. of bottom post		0.40%	3.2	Gage/4.4.5
104	Width of bridge		0.40%	3.2	Gage/4.4.5
105	True position of bridge width		0.40%	3.2	Gage/4.4.5
106	I.D. of largest center hole		0.40%	3.2	Gage/4.4.5
107	True position of largest hole		0.40%	3.2	Gage/4.4.5
108	Height from bottom to top of bridge		0.40%	3.2	Gage/4.4.5
109	Material		0.40%	3.2	Certificate
110	Distance from center line to edge of shorting bar cavity (Section B-B), 2 places		0.40%	3.2	Gage/4.4.5
111	Depth of largest hole		0.40%	3.2	Gage/4.4.5
112	Height from bottom to top of upper post		0.40%	3.2	Gage/4.4.5
113	O.D. of upper post		0.40%	3.2	Gage/4.4.5
114	True position of upper post		0.40%	3.2	Gage/4.4.5
Minor					
201	Length from bridge to closed end of battery cell cavity (2 places)		0.65%	3.2	Gage/4.4.5
202	True position of inside width of cavity (CD 203), 2 places		0.65%	3.2	Gage/4.4.5
203	Width of battery cavity (2 places)		0.65%	3.2	Gage/4.4.5

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Power Supply Molding	2	2	9292221
				NEXT HIGHER ASSEMBLY 9292241

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
204	Width of shorting bar cavity (Sect. B-B), 2 places		0.65%	3.2	Gage/4.4.5
205	Length of shorting bar slot, max. (2 places)		0.65%	3.2	Gage/4.4.5
206	Depth of shorting bar cavity (Sect. B-B), 2 places		0.65%	3.2	Gage/4.4.5
207	Width of shorting bar slot, 4 places		0.65%	3.2	Gage/4.4.5
208	Distance from center line to edge of shorting bar cavity		0.65%	3.	Gage/4.4.5
209	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

* See 4.4.5 for inspection of molding features.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63421(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Connector, Edge Clip		9292156
			NEXT HIGHER ASSEMBLY 9292158

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Solderability	20	0-1	3.2	MIL-STD-202, Method 208
Minor	None defined				

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and below.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the tables beginning with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the Contract. See section 6 of MIL-A-48078 and section 6.3 herein.

4.4.5 Dimensional control of molded parts. In place of normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1). The contractor may, with the approval of the Government, select a sample of three (3) parts (as molded) from each cavity to be fully inspected dimensionally (all dimensions on drawing) to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraph 4.4.2.1 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAA-I for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

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NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAA-M. When a new cavity is substituted, an old cavity reworked, or the material or cycle time changed, a new curing time shall be established and approved.

4.5 Test method and procedure

4.5.1 Non-activated voltage. Each battery cell of the power supply assembly shall be checked for conformance with the requirements of 3.3.1 when measured by a 10 megohm input impedance voltmeter. (Non-destructive test)

4.5.2 Shorting bar resistance. The continuity-resistance of each shorting bar circuit shall be measured, using an approved ohmmeter. Observation shall be made for compliance with requirements of 3.3.2. (Non-destructive test).

4.5.3 Battery lead weld integrity. A five (5) pound plus two (2) pound force shall be applied to each lead in a direction parallel to the lead using approved test equipment. Observation shall be made for compliance with requirements of 3.4. Visual examination shall be performed at a minimum 5X magnification. (Non-destructive test).

4.5.4 Shorting bar and connector assembly solder workmanship. Soldering and inspection shall conform to MIL-S-45743. Visual examination shall be performed at a minimum of 10X magnification. Observation shall be made for compliance with requirements of 3.5.1. (Non-destructive test).

4.5.5 Solderability. The power supply assembly leads shall be subjected to solderability procedure of MIL-S-45743. Observation shall be made for compliance with requirements of 3.5.2. (Destructive test).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The power supply assembly shall be preserved and packaged in accordance with MIL-STD-1169, Level C.

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5.2 Packing. Packing of the assemblies shall be in accordance with MIL-STD-1169, Level C.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate than the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. The component(s) covered by this specification is intended for use in FASCAM Electronic Assemblies.

6.2 Ordering data. See section 6.1 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required, to Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations of tests to Commander, ARRADCOM, ATTN: DRDAR-QAA-M, Dover, New Jersey 07801.

6.5 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, ATTN: DRDAR-QAA-M, Dover, New Jersey 07801.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1345-A134

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APPENDIX
SUPPLEMENTAL SOLDERING PROVISIONS

10 SCOPE: This appendix details supplemental soldering provisions to MIL-S-45743 for hand soldering.

20 APPLICABLE DOCUMENTS: MIL-S-45743 listed in section 2 of specification.

30 EXCEPTIONS AND CLARIFICATIONS:

30.1 Exceptions/clarifications to MIL-S-45743. The following exceptions/clarifications to reference MIL-S-45743 paragraphs shall apply.

3.3.1.2 Control of airborne contaminants. Delete the sentence, "Positive pressure shall ..... between them."

3.3.2.4.5 Soldering irons and resistance soldering irons. Delete the sentence, "The soldering iron shall ..... or standby condition.", and substitute the sentence, "The soldering iron shall be a soldering iron approved by the technical agency or a soldering iron that attains a temperature no more than 10°F (5.6C) above the control temperature when in an idling or standby condition.".

3.4.5.7 Printed board termination., Subparagraph 'b'. Change to read, " .... is permissible. Leads shall not extend, beyond the terminal pad or printed wiring pattern, far enough to cause less than .015 inches of clearance between conductors."

3.4.5.7 Printed board termination., Subparagraph 'd'. Change to read, "When leads cannot be clinched per above, they shall extend through the printed board, be visible and properly soldered."

3.4.5.8 Ribbon lead geometry. Add new sentence, "Ribbon lead termination from battery and lead assembly, shall comply with paragraph 3.4.5.7b."

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3.6.1 Cleaning the soldered connection. Change to read, " ... (within one (1) hour) using solvents or combination of solvents which will remove residual flux, activators, and other contaminants."

3.6.3 Resistivity of solvent extract. Delete.

4.2.1 One hundred percent inspection. Delete and substitute the following: "4.2.1 Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

4.2.2 Sampling inspection. Delete.

4.3.1 Visual inspection. Change to read, " ..... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

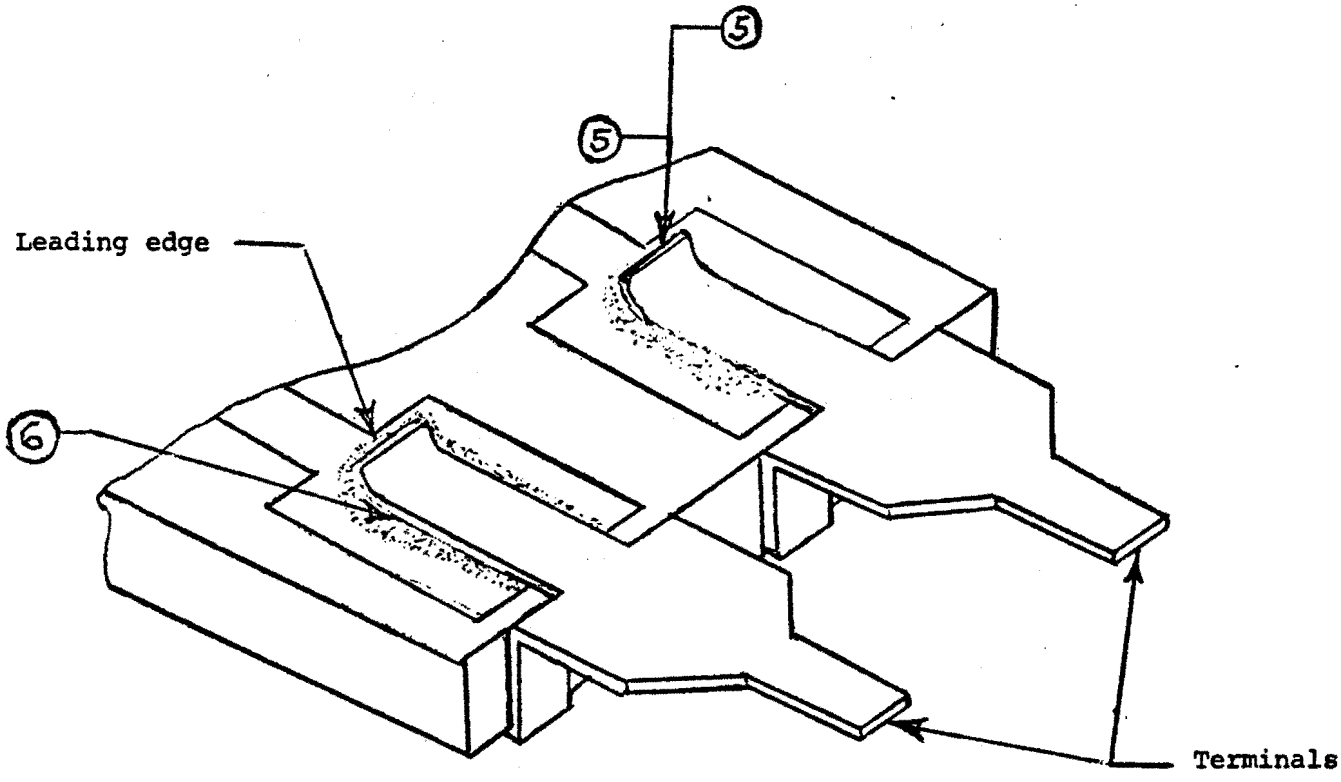
4.3.2 Cleanliness test. Delete.

40 SUPPLEMENTAL SOLDER WORKMANSHIP STANDARDS

40.1 General. The following solder workmanship standards supplement the workmanship requirements contained in MIL-S-45743 and are provided as additional aids in establishing the minimum acceptance standards for soldering. All examinations will be performed using a minimum 5 diopter lens. Suspected connections may be examined using 10X magnification to determine acceptablility.

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40.2 Shorting Bar Soldering. This section describes the acceptance criteria for solder connections of terminal to pad on ceramic substrate. All examinations shall be performed with 10x magnification (Microscope).



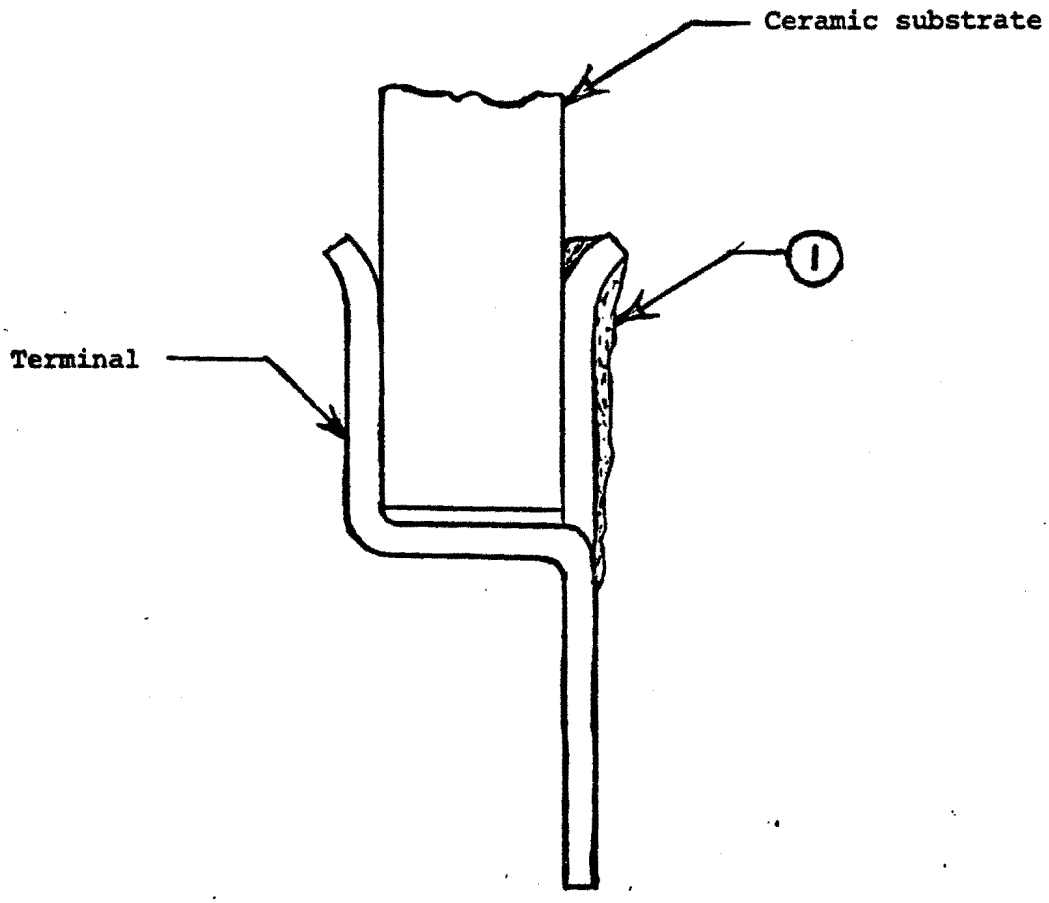
PREFERRED

1. Terminals are straight, parallel to and centered on solder pad.
2. Solder fillets are continuous on all edges, for the entire length of terminal/pad interface without voids.
3. Solder wetting is evident on terminal edges and solder pad.

ACCEPTABLE

4. Solder is continuous and wetted on the entire leading edge and along both sides for 50% of the terminal/pad interface. No cracks or voids may appear between terminal and solder pad in the soldered area.
5. Solder is continuous and wetted on the leading edge and one (1) of two (2) edges of terminal for the entire length of the solder pad/terminal interface. No cracks or voids may appear between terminal and solder pad in soldered area. There must be evidence of some solder wetting on opposite edge.
6. Solder fillet, 3/4 thickness of terminal, minimum.

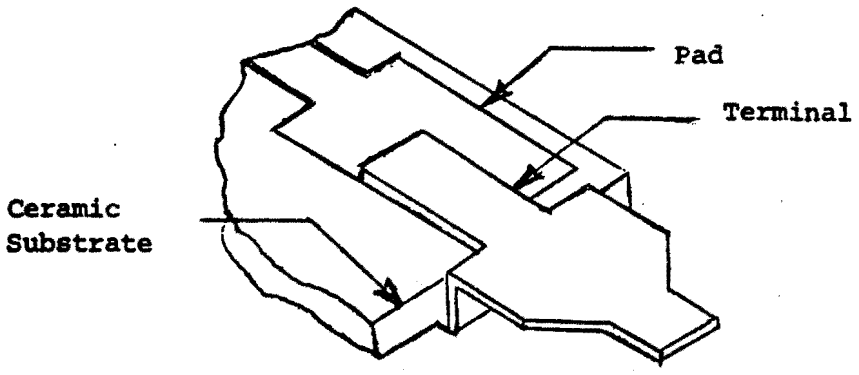
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ACCEPTABLE

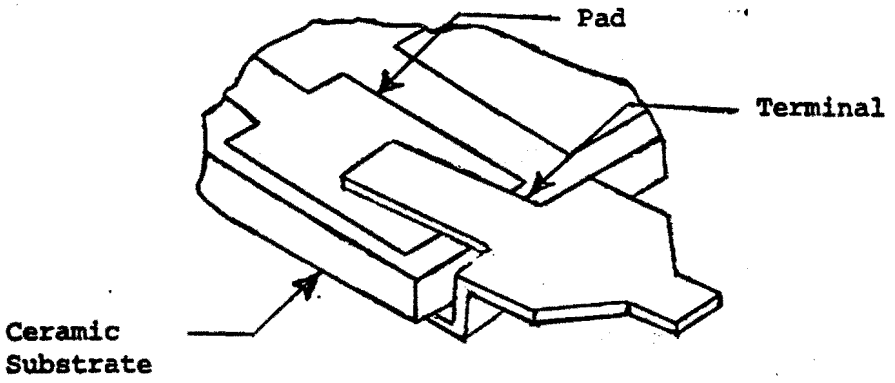
1. Solder buildup is not greater than one (1) terminal thickness.

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ACCEPTABLE MINIMUM

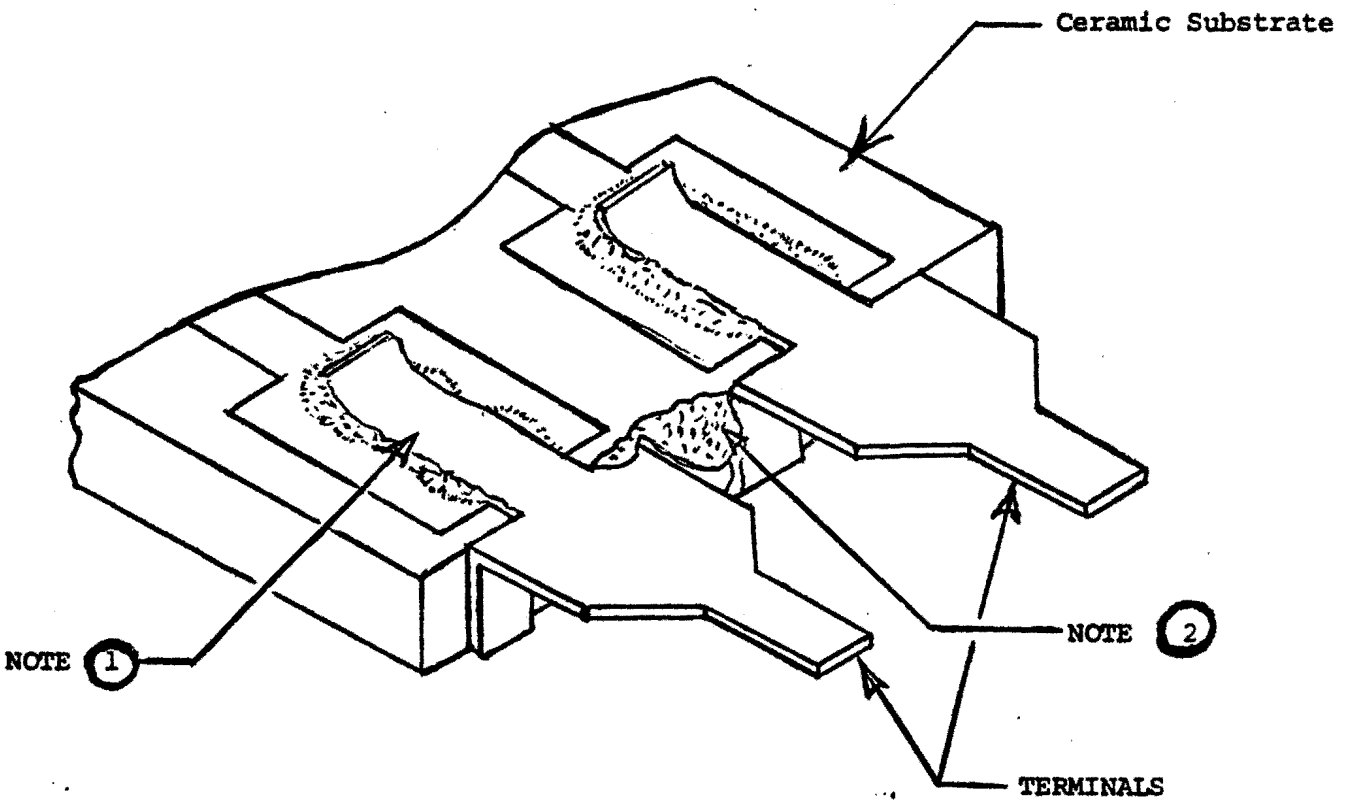
Terminal edge coincident with pad edge



ACCEPTABLE MINIMUM

Terminal slightly skewed, but does not extend beyond pad edge

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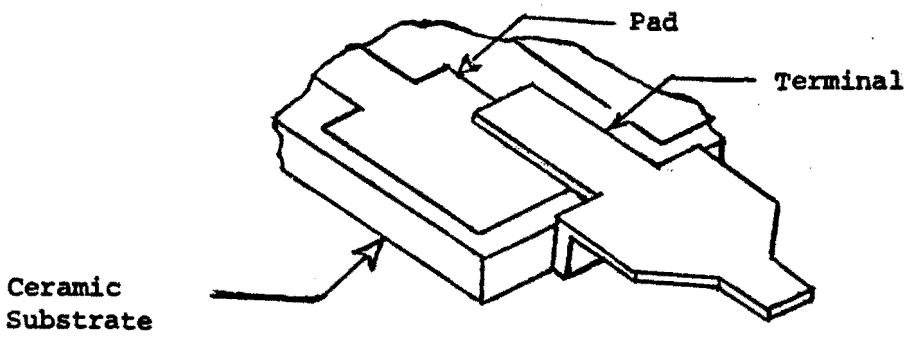


UNACCEPTABLE

NOTES:

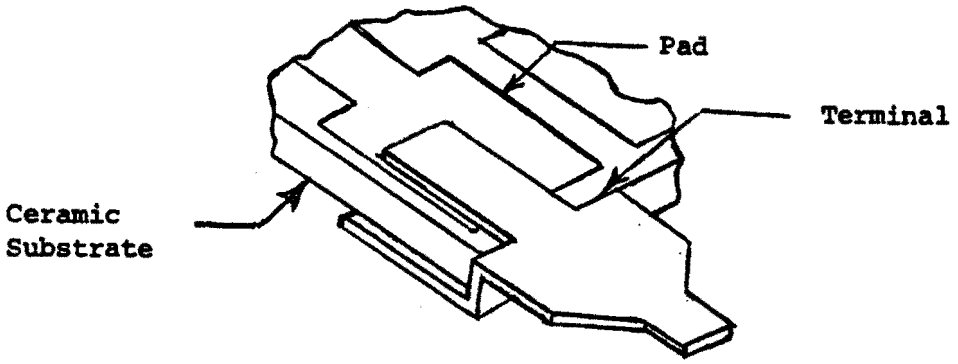
1. **Gaps in solder fillet exceed Note 4 of figure shown in paragraph 40.**
2. **Bridging between terminals**

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UNACCEPTABLE

TERMINAL OVERLAPS PAD



UNACCEPTABLE

Terminal overlaps ceramic substrate

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MIL-P-63421 Power supply-assemblies for FASCAM electronic assemblies

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